DARSHITH RAJ KATEPOGU

AI / ML Engineer | Generative AI Associate | Data Scientist

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AI & Data professional with 2.5 years of experience building machine learning models, automation pipelines, and analytics-driven solutions across India and Japan. Strong foundation in Python, SQL, ML modeling, and data workflows with hands-on experience in API integrations and workflow automation. Currently expanding into Generative AI systems including LLM-powered applications, RAG pipelines, and semantic search solutions. Passionate about building scalable, data-driven and AI-powered business solutions.

Professional Objective

- Applied predictive analytics and machine learning techniques using Scikit-learn to support data-driven business decision-making and performance forecasting.
- Designed and automated data workflows through Python-based ETL pipelines and API integrations to streamline data collection, transformation, and processing.
- Built interactive, data-driven dashboards and analytical applications using Streamlit and Power BI to enable stakeholders to explore insights in real time.
- Developed intelligent document retrieval and semantic search systems using LangChain, vector embeddings, and FAISS to enable efficient knowledge discovery across structured and unstructured data sources.
- Implemented Retrieval-Augmented Generation (RAG) architectures by integrating OpenAI APIs with vector search to deliver context-aware and accurate AI-generated responses.
- Supported structured data analysis and reporting through advanced SQL queries, joins, aggregations, and performance tuning across relational databases.
- Contributed to scalable AI solutions by integrating ML models into REST APIs for seamless application-level deployment and enterprise consumption.
- Enhanced data processing efficiency by automating repetitive workflows and validation tasks using Python scripts and modular pipelines.
- Collaborated in building end-to-end analytics solutions covering data ingestion, model development, evaluation, deployment, and monitoring.
- Explored and implemented Generative AI use cases, including prompt engineering, embeddings optimization, and reducing hallucinations through grounded retrieval techniques.
- Worked closely with cross-functional teams in Agile sprints to iterate on models, improve solution performance, and deliver production-ready AI workflows across ML, NLP, and application layers.

Domain Expertise

✓ Customer Analytics & Prediction
✓ Workflow Automation
✓ Business Reporting & Dashboarding
✓ Document Intelligence Systems (AI-based Retrieval)
✓ Data Processing & Transformation
✓ Data-driven Reporting & KPI Analysis
✓ Cross-functional Collaboration for Data Insights
✓ Real-world ML deployment exposure across India and Japan environments

Education

Bachelors of technology - 2023 from CMR College of Engineering and Technology (CMRCET),Hyderabad,Telangana,India
Data Science Certification - (6 months) Trained from ExcelR institute , Hyderabad,Telangana,India

Core Competencies

Programming	Python
Data Analytics	EDA, Pandas, NumPy, Data Cleaning, Data Transformation
Machine Learning	Supervised and Unsupervised learning,Classification, Regression, Feature Engineering, Model Evaluation, Scikit-learn.
Data Engineering	ETL Pipelines, Workflow Automation, API Integration
Natural Language Processing	Text Preprocessing, Tokenization, Semantic Search, Text Classification
SOA	FastAPI (Learning), Streamlit, REST API Integration
Generative AI & Machine Learning	Core Concepts: NLP, LLMs, RAG (Retrieval-Augmented Generation), Prompt Engineering, Embeddings. Models & Architectures: GPT-3/4, PaLM, LLaMA, BERT, GANs, VAEs, Transformers, Stable Diffusion. Frameworks & Libraries: LangChain, LangGraph, Hugging Face Transformers, TensorFlow, Keras, PyTorch, scikit-learn. Data & Processing: Pandas, NumPy, OpenCV. Agentic & Workflow Systems: PhiData Agents. AI-Powered Search: Vector Search, Semantic Search, AI-powered Web Search (Google, DuckDuckGo) Vector Databases: ChromaDB, FAISS, Pinecone
Visualization & Apps	Streamlit, Power BI (Learning)
SQL & NoSQL	Microsoft SQL Server, MongoDB.
Markup Languages	HTML, HTML5, CSS, XML.
Client Scripting Languages & CSS	Java Script, AJAX and jQuery, JSON
DevOps Pipeline	GitHub, GitLab, and GitHub Actions
API Testing	Postman, SoapUI, ReadyAPI, Bruno Tools
SDLC methodologies	Agile, Waterfall, DevOps, Spiral, Iterative and V-Model.
ALM (Lifecycle)	Atlassian Jira and Rally -CA Agile Central

Experience

eM Cube Corporation — **Tokyo,Japan**
Associate Data Scientist (Contractual) — **08/2025 - 02/2026**

- Worked on building AI-driven analytical solutions using real-world customer and operational datasets to identify patterns, trends, and actionable insights for business teams.
- Developed machine learning models using Python and Scikit-learn, handling the full process from data cleaning and feature engineering to training, tuning, and evaluating model performance.
- Designed and implemented a Retrieval-Augmented Generation (RAG)–based solution to improve the accuracy of LLM responses by connecting the model with enterprise data sources instead of relying only on base model knowledge.
- Built an LLM-powered Q&A assistant using LangChain, embeddings, and vector search to retrieve relevant information from structured databases and unstructured documents such as PDFs and knowledge pages.
- Created document ingestion pipelines that included text chunking, embedding generation, and indexing into a vector database to enable efficient semantic search.
- Applied hybrid search techniques (keyword + vector similarity) to improve retrieval relevance and reduce hallucinations in generated responses.
- Worked on data preprocessing, feature selection, and validation techniques to improve prediction reliability and

overall model performance.

- Used SQL regularly for data extraction, validation, and exploratory analysis to support model development.
- Developed simple Streamlit-based prototypes to demonstrate ML models and RAG solutions for internal stakeholders and collect feedback.
- Worked in an Agile team, participating in sprint planning, experimentation cycles, and preparing AI solutions for deployment.

Environment: Python, Scikit-learn, LangChain, SQL, NLP techniques, Streamlit, RAG.

Swift Solutions Pvt Ltd **06/2024 - 07/2025**
Data Analyst, Bengaluru, India

- Developed analytical reports and dashboards to support business decision-making through structured data insights.
- Designed SQL-based data extraction queries to pull and transform business-critical data from relational databases.
- Built Python-based ETL workflows to automate data cleaning and preparation processes.
- Conducted exploratory data analysis (EDA) to identify patterns and performance gaps.
- Assisted in implementing classification and regression models to support operational forecasting.
- Delivered data visualization solutions for stakeholder reporting using Python-based tools.
- Worked closely with cross-functional teams to understand data requirements and translate them into analytical solutions.

Environment: SQL, Python, Pandas, Machine Learning basics, Data Visualization

AI Variant, Hyderabad, India **04/2023 - 04/2024**
Data Science Intern

- Supported the development of machine learning models for classification and predictive analysis tasks.
- Worked on text preprocessing and NLP workflows including tokenization and feature extraction.
- Assisted in building a semantic search prototype using embeddings for improved information retrieval.
- Performed data cleaning, transformation, and exploratory analysis using Python.
- Participated in model training and evaluation using Scikit-learn.
- Contributed to the development of small-scale AI solutions demonstrating real-world use cases.

Environment: Python, Scikit-learn, NLP techniques, Data Preprocessing